PROJECT PROPOSAL

***Do not exceed 2 pages**

1. PROJECT TITLE	**AquaTrace – Community- Verified Water Pipe Repair Tracker**	
2. DATE	**23rd September, 2026**	
3. PREPARED BY	**NAME:** N P A Sasundi Binara G S Vithanage J C Jayalakshi W M S S Wijekoon I G R M Dharmappriya J K N S Jayakody M D A H Mudalige B M D Chamindi	**INDEX:** 33864 34935 34724 34920 35500 34954 35535 35359
4. SUPERVISOR NAME	Mr. Diluka Wijesinghe	

5. PROBLEM STATEMENT

The cases where water pipes have broken are usually late reports or unaudited reports where the faults are not well known; hence, road excavation becomes necessary. Once repair has been completed, there is no means through which one can check the effectiveness of the repair as the fault might still be present until the next report is made.

6. PROPOSED SOLUTION

In AquaTrace, all these three groups are linked within a common map-based framework. The citizen files the report together with the photograph and the coordinates while the administrator authenticates the report and assigns the technician who in turn makes the repair and marks the task as completed. This then leads to confirmation by other citizens in the vicinity of the job.

7. OBJECTIVES & DELIVERABLES

- Fault reporting which is efficient and precise with photographic and location evidence.
- Process flow to ensure that the faults are assigned and repaired in a proper manner.
- Citizens verifying repair processes.
- Flutter app which incorporates databases in the cloud with role-based authentication and Git commit history.

8. TARGET USERS

Citizen/Public User — reports faults, verify repairs
Water Board Technician — receives jobs, updates status
Water Board Admin/Dispatcher — verifies, assigns, monitors

9. CORE FEATURES

- Fault Reporting — Users can report water pipe faults with photos captured using the camera, automatically recorded GPS coordinates, a description, and severity level
- Interactive Fault Map — Displays all reported faults on a map using different colors for each status: pending, assigned, ongoing, completed, and disputed
- Admin Dashboard — Allows administrators to review and verify submitted reports, assign verified faults to technicians, and monitor repair progress
- Technician Job Queue — Technicians receive assigned jobs and navigate directly to the exact fault location
- Citizen Verification — Nearby users (within a defined radius) can confirm a repair is fixed or dispute it if the issue persists
- Status Notifications — Sends notifications when important changes occur, such as a new technician assignment, report status update, or dispute
- Report History — Maintains a complete status and activity history for each fault from initial submission through final resolution

10. TECHNOLOGIES TO BE USED

Frontend: Flutter (Dart)
Backend: Firebase
Database: Firebase Cloud Firestore
Authentication: Firebase Authentication
File Storage: Firebase Storage
Notifications: Firebase Cloud Messaging (FCM)
Maps: OpenStreetMap
Map Integration: flutter map
Location Services: Geolocator
Camera: Flutter Camera / Image Picker
Version Control: GitHub
IDE: Android Studio / VS Code
Cost: Free / No-cost for coursework and demonstration within free usage limits.

11. NOVELTY

As opposed to traditional methods that only report and track a repair, the AquaTrace system allows for independent verification by citizens after the completion of the repair, and the system will automatically re-open the case if the solution does not hold.

12. TIMELINE & MILESTONES

Phase	Milestone / Task	Estimated Completion
Phase 1	Proposal approval, requirements finalization, UI/UX design	10 Oct 2026
Phase 2	Database setup, core reporting & map features	27 Oct 2026
Phase 3	Job assignment workflow, citizen verification, notifications	13 Nov 2026
Phase 4	Testing, bug fixing, report writing, final submission & demo	28 Nov 2026